

Exp No: 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

PROGRAM:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int i;
```

```
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
```

```
    unsigned long sum = 0;
```

```
    for (i = 0; i < 5; i++)
```

```
    {
```

```
        sum = sum + array[i];
```

```
    }
```

```
    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
```

```
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
```

```
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
```

```
    while(1); // keep output stable
```

```
}
```

OUTPUT:

The screenshot displays the Keil uVision IDE interface during a simulation. The main window shows the C source code for `exp13.c`, which includes a `while(1)` loop. The `Registers` window on the left shows the state of various registers, including `R0` through `R15`. The `Disassembly` window shows the assembly code corresponding to the C code. The `Command` window at the bottom left shows the command `Load "C:\\Keil_v5\\CS1\\Examples\\HELLO\\Objects\\exp13"`. The `Call Stack - Locals` window on the right shows the current function `MAIN` and its local variables `i`, `array`, and `sum`. Three floating windows show the status of Parallel Ports 0, 1, and 2, with their respective data and pins.

Registers

Register	Value
R0	0x00
R1	0x08
R2	0x00
R3	0x05
R4	0x00
R5	0x00
R6	0x00
R7	0x01
R8	0x01
R9	0x08
R10	0x15
R11	0x17
R12	0x00
R13	0x00
R14	0x00
R15	0x00
PC	0x0979
SP	0x0979
SR	0x00000000
PSW	0x01

Disassembly

```
18: while(1);
C:0x0979 80FE SJMP C:0979
133: MOV R0,#IDATALEN - 1
C:0x097B 787F MOV R0,#0x7F
```

exp13.c

```
1 #include <reg51.h>
2
3 void main(void)
4 {
5     unsigned int i;
6     unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
7     unsigned long sum = 0;
8     for (i = 0; i < 5; i++)
9     {
10        sum = sum + array[i];
11    }
12    P0 = (unsigned char)(sum & 0xFF);
13    P1 = (unsigned char)((sum >> 8) & 0xFF);
14    P2 = (unsigned char)((sum >> 16) & 0xFF);
15    while(1);
16 }
```

Parallel Port 2

Port 2	7	6	5	4	3	2	1	0
P2	0x01							
Pins	0x01							

Parallel Port 1

Port 1	7	6	5	4	3	2	1	0
P1	0xAB							
Pins	0xAB							

Parallel Port 0

Port 0	7	6	5	4	3	2	1	0
P0	0xCC							
Pins	0xCC							

Command

```
Running with Code Size Limit: 2K
Load "C:\\Keil_v5\\CS1\\Examples\\HELLO\\Objects\\exp13"
```

Call Stack - Locals

Name	Location/Value	Type
MAIN	C:0x0909	
i	0x0005	uint
array	D:0x08	array[5] of uint
sum	0x0001ABCC	ulong

Simulation

t1: 0.00055600 sec L18 C1 CAP NUM SCRL OVR R/V

31°C Light rain

1800 07-08-2026